

Multiple Choice

1. **Answer:** D

Options: A) A college buys computers.; B) The Department of Transportation repaves a road.; C) A city spends money to refurbish a public park.; D) You buy a 1957 Chevy from a friend.

Note: The purchase of a 1957 Chevy from a friend does not affect GDP because it is a transaction of a used good, which does not contribute to current economic production measured by GDP.

2. **Answer:** E

Options: A) A wage payment lower than the average revenue product of labor.; B) Employment level lower than that of a competitive labor market.; C) A single buyer of labor services.; D) Employment level greater than that of a competitive labor market.; E) A wage payment lower than the marginal revenue product of labor.

Note: A monopsonist faces downward-sloping labor supply and thus pays a wage that is less than the marginal revenue product of labor, maximizing its profit while minimizing labor costs.

3. **Answer:** B

Options: A) add \$1000 to the money supply; B) add \$ 500 to the money supply; C) no change in the money supply; D) reduce \$1000 from the money supply

Note: The correct answer is due to the fact that when Bank A receives a 1,000 deposit, it increases the loan to Mr. Y, it effectively creates new money in the banking system, resulting in a net increase of \$500 in the overall money supply.

4. **Answer:** D

Options: A) the demand for goods and services decreases.; B) the stock market crashes.; C) bond prices fall.; D) the nominal interest rate falls.; E) the unemployment rate rises.

Note: Households demand more money as an asset when the nominal interest rate falls because lower interest rates reduce the opportunity cost of holding money, making it more attractive compared to other interest-earning assets.

5. **Answer:** A

Options: A) \$50 billion; B) \$87.5 billion; C) \$100 billion; D) \$37.5 billion; E) \$200 billion

Note: The correct answer is \$50 billion because a lump-sum tax directly reduces disposable income by that amount, and with a marginal propensity to save (MPS) of 0.25, the entire reduction in disposable income leads to an equivalent decrease in net national product (NNP) without any multiplier effect from consumption.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because $\hat{u}$ represents a vector of estimated residuals in regression analysis, and the product $\hat{u}^t \hat{u}$ results in a scalar value, which has dimensions of 1×1 .

7. **Answer:** False

Options: A) True; B) False

Note: Crowding out is more likely in a boom period because higher demand for investment can lead to increased interest rates, which in turn reduces private investment, even if the investment demand curve is perfectly inelastic.

8. **Answer:** False

Options: A) True; B) False

Note: A minimum wage in the market for fast-food workers is unlikely to create a shortage of available positions because it can lead to increased worker retention and productivity, as well as attract a larger pool of applicants, ultimately ensuring that businesses fill their positions rather than reducing them.

9. **Answer:** True

Options: A) True; B) False

Note: A value-added tax is indeed collected at each stage of production because it is levied on the increase in value that a product undergoes as it moves through the production process, ensuring that tax is applied progressively based on the contributions made at each stage.

10. **Answer:** False

Options: A) True; B) False

Note: Lenin's economic theory included the establishment of state welfare mechanisms as part of the transition from capitalism to socialism, emphasizing the importance of state intervention in the economy to support the working class.

Long Form Response

11. **Answer:** When the Federal Reserve increases bank reserves by 500,000 *with a required reserve ratio of*

Note: correct because it accurately applies the money multiplier concept to demonstrate that the Federal Reserve

11. **Answer:** Higher interest rates can significantly mitigate the effectiveness of expansionary fiscal policy by discouraging private investment. When the government increases spending or cuts taxes to stimulate the economy, the resulting higher demand can lead to increased borrowing, which tends to push interest rates upward. As borrowing becomes more expensive, businesses may delay or reduce their capital expenditures, negatively impacting private investment. This reduction in private investment can

stifle economic growth, as it limits the potential for increased productivity and innovation. Consequently, the intended stimulative effects of fiscal policy may be undermined, leading to a less robust economic recovery than anticipated.

Note: correct because it accurately identifies the relationship between increased government spending, rising interest rates, and the subsequent discouragement of private investment, illustrating how these dynamics can hinder overall economic growth.